

High-frequency electromagnetic wave propagation in dispersive bianisotropic media

Éric Savin^{1,*}, Jean-Luc Akian²

¹Computer Science Department, ONERA, Palaiseau, France

²Materials and Structures Department, ONERA, Châtillon, France

*Email: Eric.Savin@onera.fr

Abstract

We present a mathematical model for the propagation of high-frequency electromagnetic waves in a randomly heterogeneous bianisotropic dielectric medium. High frequencies correspond to the situation where the wavelength is comparable to the correlation length of random inhomogeneities, when strong interactions occur between the waves and the propagation medium. We establish the transport and radiative transfer equations for the electromagnetic energy densities associated to the different wave polarizations. Our analysis extends existing results to account for the heterogeneity of the background propagation medium and possible coupling between the different polarizations. It is based on the interpretation of Wigner transforms and Wigner measures (characterizing the energy densities in phase space) of electromagnetic waves in terms of semi-classical pseudo-differential operators in their standard quantization.

Keywords: Maxwell's equations, bianisotropic dielectric media, kinetic model, transport equation, radiative transfer.

1 Introduction

The analysis of the propagation of electromagnetic waves in complex media is of interest for a wide range of applications, including radar imaging and remote sensing, optical imaging, atmospheric physics, or communications among others. As the complexity of the propagation media cannot be described exactly but only evaluated in a statistical sense, we model their inhomogeneities (or imperfections) as random fields and thus consider Maxwell's equations with randomly fluctuating electromagnetic parameters:

$$\partial_t \begin{bmatrix} \varepsilon & \xi \\ \xi^* & \mu \end{bmatrix} \begin{pmatrix} E \\ H \end{pmatrix} + \nabla \times \begin{bmatrix} 0 & -I_3 \\ I_3 & 0 \end{bmatrix} \begin{pmatrix} E \\ H \end{pmatrix} = 0, \quad (1)$$

where E is the electric field, H is the magnetic field, and I_d is the $d \times d$ identity matrix. The electromagnetic parameters which vary ran-

domly with position x are the permittivity tensor $\varepsilon(x)$, the permeability tensor $\mu(x)$, and the magnetoelectric tensor $\xi(x)$, for $\xi^*(x)$ being its conjugate transpose.

Asymptotic methods can be used to characterize wave fields both quantitatively and qualitatively when some explicit scales can be identified and separated to describe the corresponding propagation regimes. Recent mathematical works on electromagnetic waves in random media have considered stratified half-spaces [5], waveguides [2], or open environments [3], for example. In these works, a preferred direction of propagation can be identified and the electromagnetic fields can be decomposed into plane waves transverse to this direction. Their amplitudes are random fields of which evolution is driven by random fluctuations of the dielectric parameters, and can be studied in different asymptotic regimes. For example, cumulative scattering effects induced by small inhomogeneities at large propagation distances in a waveguide are generally manifested by the exponential decay of the mean fields, the increase of their random fluctuations, and depolarization.

2 Results

In our research, we rather study the situation in which no preferred propagation direction can be identified beforehand, as the waves propagate in open random media. The wavelength and correlation length of random fluctuations of electromagnetic parameters are small and comparable, while the propagation distance and/or observation time are large. The amplitude of random fluctuations is also small. The random electromagnetic parameters are then written:

$$\begin{aligned} \begin{bmatrix} \varepsilon & \xi \\ \xi^* & \mu \end{bmatrix} (x) &= K(x) \\ &= K_0(x) \left[I_6 + \sqrt{\epsilon} V \left(\frac{x}{\epsilon} \right) \right] \end{aligned} \quad (2)$$

where $\epsilon > 0$ is the small correlation and wavelength scale, K_0 are the mean background parameters, and V are their random fluctuations.

This high-frequency scaling is relevant if the objective is, for example, to image inhomogeneities from remote sensors. In this context, waves are scattered by inhomogeneities as they propagate in all directions, and their polarization and phase are randomized. We then know that the energy density of the waves is the relevant quantity to focus on. It can be evaluated from the Wigner transform W_ϵ of the wavefield $u_\epsilon = (E, H)$ solving (1) with (2):

$$W_\epsilon(x, k) = \int e^{ik \cdot y} u_\epsilon(x - \epsilon y) u_\epsilon^*(x) \frac{dy}{(2\pi)^3} \quad (3)$$

and its high-frequency limit $W = \lim_{\epsilon \rightarrow 0} W_\epsilon$, the associated Wigner measure which is a positive quantity characterizing the energy density in phase space [6]. The evolution of the Wigner measure W is described by coupled radiative transfer equations for the different polarization of the waves. Their collisional operators are linear with kernels depending on the correlation structure of the random fluctuations V [7]. In fact, the sole information needed to account for the fluctuations of electromagnetic parameters is their power spectral densities, *i.e.* the spatial Fourier transforms of their correlation functions. Radiative transfer equations for high-frequency electromagnetic vector waves have already been obtained in [7] for random fluctuations in isotropic media, and in [4] for random fluctuations in non-dissipative bianisotropic media with constant background parameters K_0 . We have extended these results to very general constitutive equations in [1], including heterogeneous materials with dispersive and dissipative properties such that K in (2) depends on both position and time. For that purpose we relied on the interpretation of Wigner transforms (3) in terms of semi-classical pseudo-differential operators in space and time.

3 Outlook

Radiative transfer equations have a direct geometrical interpretation in terms of bicharacteristic rays, *i.e.* rays in phase space, which are similar to the rays of classical ray methods (or geometric optics). This approach is applicable to the simulation of synthetic aperture radar (SAR) for example, including the consideration of multiple paths in radar imaging, remote sensing, or for the calculation of antenna transfer functions in heterogeneous environments when-

ever the approximations of geometric optics hold. Ray tracing solvers are then required, such as those that have been developed in recent years in computer graphics for example. One of the possible applications of this work concerns the extension of these solvers for ray tracing in phase space and in complex media such as plasmas. Algorithms and applications implemented in the video game industry, among others, could thus be reused in the context of numerical electromagnetism.

References

- [1] J.-L. Akian and É. Savin, Wigner measures of electromagnetic waves in heterogeneous bianisotropic media, *Wave Motion* **127** (2024), 103296.
- [2] R. Alonso and L. Borcea, Electromagnetic wave propagation in random waveguides, *SIAM Multiscale Model. Simul.* **13** (2015), pp. 847–889.
- [3] L. Borcea and J. Garnier, Polarization effects for electromagnetic wave propagation in random media, *Wave Motion* **63** (2016), pp. 179–208.
- [4] A. Fannjiang, Two-frequency radiative transfer: Maxwell equations in random dielectrics, *J. Opt. Soc. Am. A* **24** (2007), pp. 3680–3690.
- [5] J. Garnier and K. Sølna, Transmission and reflection of electromagnetic waves in randomly layered media, *Commun. Math. Sci.* **13** (2015), pp. 707–728.
- [6] P. Gérard, P. A. Markowich, N. J. Mauser, and F. Poupaud, Homogenization limits and Wigner transforms, *Comm. Pure Appl. Math.* **L** (1997), pp. 323–379.
- [7] L. V. Ryzhik, G. C. Papanicolaou, and J. B. Keller, Transport equations for elastic and other waves in random media, *Wave Motion* **24** (1996), pp. 327–370.